Question:

An isosceles right triangle with legs of length 8 is partitioned into 16 congruent triangles as shown. The shaded area is

$\text{(A)}\ 10 \qquad \text{(B)}\ 20 \qquad \text{(C)}\ 32 \qquad \text{(D)}\ 40 \qquad \text{(E)}\ 64$

Reasoning + Answer:

<think> To find the shaded area in the given isosceles right triangle, we start by calculating the area of the entire triangle. The legs of the isosceles right triangle are each 8 units long. The area \(A \) of a right triangle is given by the formula:

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

Here, the base and height are both 8 units. So the area of the entire triangle is:

$$A = \frac{1}{2} \times 8 \times 8 = 32 \text{ square units}$$

The triangle is partitioned into 16 smaller triangles, which means each smaller triangle has an area of:

$$\text{Area of each smaller triangle} = \frac{32}{16} = 2 \text{ square units}$$

Now, we need to determine how many of these smaller triangles are shaded. From the diagram, we can see that the shaded area is composed of 10 of the smaller triangles. Therefore, the shaded area is:

$$\text{Shaded area} = 10 \times 2 = 20 \text{ square units}$$

Thus, the shaded area is 20 square units. </think>

<answer>20</answer>